Harris Corner Detector

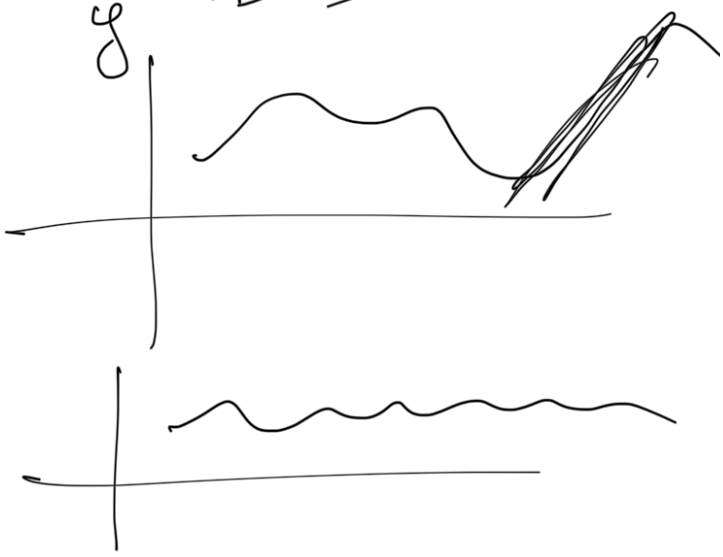
Used to find Corners in an image.

Definition of Corner

Sharp change in underlying value.



1D ⇒



In 2D we ~~~~ find sharp changes using gradients (or derivatives)

Have to find a method to evaluate derivatives in an image

Sobel filtering

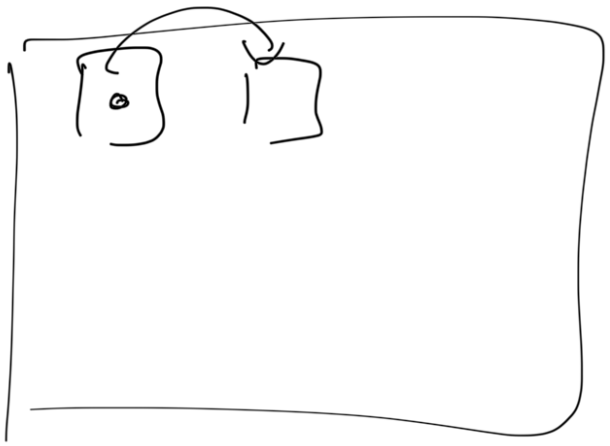
$$\frac{df}{dx} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$I_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}_{3\times 3}$$

Sobel filter in x-direction

→ It gives derivatives in x-direction

Constant region & result of filtering is zero.



filtered image has a value iff there is a sharp in x-direction

Gradient in y-direction

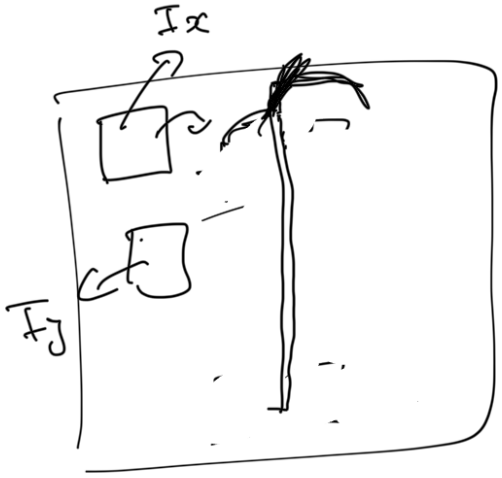
$$I_y = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

$$I = \sqrt{I_x^2 + I_y^2}$$

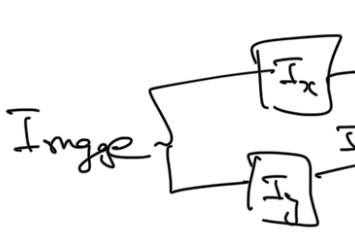
Overall gradient

$I_x \Rightarrow$ defect edges in x-direction

$I_y \Rightarrow$ defect edges in y-direction



Corners $\Rightarrow$ Spatial points at which there is an edge in both x and y-direction



CV wise

Both I_x and I_y image have high values at these points.

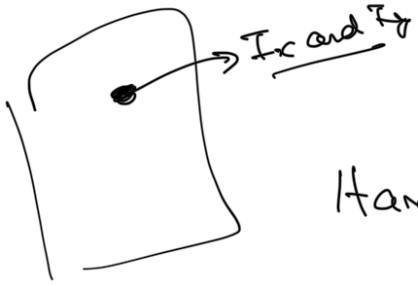


Why is Sobel insufficient for robust Corner detection :

edge

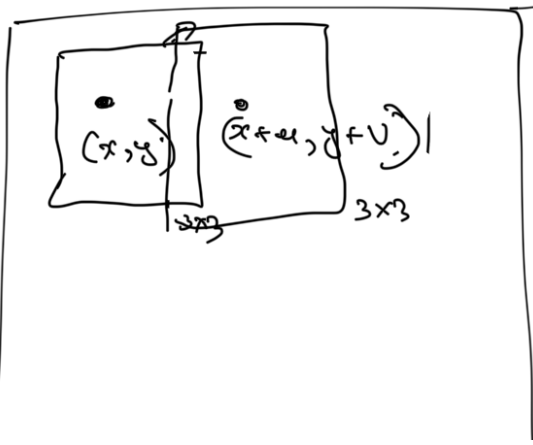
1) You have to find a threshold
↓
threshold changes from application to application

2) Sobel is effected by noise
↓
not robust to noise.

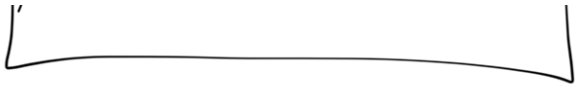


Harris Corner method
↓
gives you a new scalar (R) value which we can use to find corners.
↓
generates R for every pixel
↓
wherever R > some value
it's point of corner in 2D grid.

W ⇒ [ ]$_{w \times w}$
2×2
↓
box size
↓
corners or edges will classified as edges or corners if they



area at least 2 pixel thick.



$w(i,j)$ = 2D Gaussian

Kernel size => 3x3
&
Kernel size for Sobel filtering

Squared error

$$E_{u,v}(x,y) = \sum_{j=-1}^{1}\sum_{i=-1}^{1} w(i,j)\left(I(x+u+i, y+v+j) - I(x+i, y+j)\right)^2$$

$$I(x+u+i, y+v+j) \approx I(x+i, y+j) + I_x u + I_y v + \ldots \quad \text{---(1)}$$

Taylor series in 1D

$$f(x+\Delta x) = f(x) + f'(x)\Delta x + f''(x)(\Delta x)^2 + \ldots$$

Using (1)

$$E_{u,v}(x,y) = \sum_{j=-1}^{1}\sum_{i=-1}^{1} w(i,j)\left(I_x u + I_y v\right)^2$$

$E_{uv}(x,y)$ is high for edges
$E_{uv}(x,y)$ is high for Corners

Quadratic

$$E_{uv}(x,y) = \sum_{j=-1}^{1}\sum_{i=-1}^{1} \Big(w(i,j)\, I_x^2\, u^2 + w(i,j)\, I_y^2\, v^2 + 2\, w(i,j)\, I_x I_y\, uv \Big)$$

In Matrix form

$$E_{uv}(x,y) = \underset{1\times 2}{[u \;\; v]} \begin{bmatrix} A & B \\ B & C \end{bmatrix}_{2\times 2} \begin{bmatrix} u \\ v \end{bmatrix}_{2\times 1}$$

$$A = \sum_{j=-1}^{1}\sum_{i=-1}^{1} w(i,j)\, I_x^2 \qquad B = \sum_{j=-1}^{1}\sum_{i=-1}^{1} w(i,j)\, I_x I_y$$

$$C = \sum_{j=-1}^{1}\sum_{i=-1}^{1} w(i,j)\, I_y^2$$

$I_x \Rightarrow$ Img filtered using sobel filter $\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$

$I_y \Rightarrow$ Img filtered using sobel filter $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$

$E_{uv}(x,y)$ — If it increases in both u and v, it is point of a corner

$E_{uv}(x,y)$ — If it increases only in

the direction of
if point of
edge.

$$M = \begin{bmatrix} A & B \\ B & C \end{bmatrix}$$

Eigen Decomposition

$$M = E \downarrow E^T$$

eigen values

$$\begin{bmatrix} A & B \\ B & C \end{bmatrix} = [e_1\ e_2] \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

$e_1\lambda_1^2 + e_2\lambda_2^2$

$E(e_1, e_2)$: (Change of coordinate)

Corner — If $E(u)$ is increasing in both e_1 and e_2 direction.

↓

Both the eigen values are high.

$E \in (0.04, 0.06)$

$\lambda_1, \lambda_2 \Rightarrow$ High $\Rightarrow$ Corner

λ_1 high, λ_2 low $\Rightarrow$ edge

λ_1 low, λ_2 high $\Rightarrow$ edge

$k = 0.04$

Constant to use in Harris Corners

$$|M| = \lambda_1 \lambda_2$$

$$\text{Trace}(M) = \lambda_1 + \lambda_2$$

$$R = |M| - k(\text{trace}(M))^2$$

R	λ_1, λ_2	Edge or Corner
$R > 0$	λ_1 high, λ_2 high	Corner
$R < 0$	λ_1 high, λ_2 low	Edge
$R < 0$	λ_2 high, λ_1 low	Edge